



NF D-42

Digital delay processor

User manual



NF Audio Tools by Nenno Fernando

Version 1.0.0

VST3 - Audio Unit - AAX

Introduction

NF D-42 is a digital delay processor with a long delay memory, a voltage-controlled sweep (VCO), a programmable clock and an infinite-repeat function. It covers everything from tight doubling, chorus and flanging (a few milliseconds) through slap-back and tape-style echoes, to long delays and live looping of up to 2.4 seconds.

It is modelled on the behaviour described in the public owner's manual of a classic 1U rack digital delay: delay ranges, sweep ratio, input limiter, filters, clock and repeat logic follow that manual. It is an original NF Audio Tools product; it is not affiliated with, or endorsed by, the maker of any hardware it was inspired by, and all trademarks belong to their respective owners.

Quick start

- Insert NF D-42 on a track or aux (stereo or mono). Pick a preset from the preset tab (top right), for example "Slapback" or "Flanger".
- Set LEVEL so the HEADROOM lamps reach the amber lamp (-6) on peaks. The red lamp (0 dB) means the input limiter is working.
- OUTPUT MIX blends the direct and the delayed signal (centre = equal blend). FEEDBACK sets how many repeats you hear.
- Set the delay time with the UP / DOWN buttons, the mouse wheel over the display, or by dragging the display up and down.
- Add movement with the VCO-SWEEP section: DEPTH for the amount, RATE for the speed and WAVEFORM for the shape.

The front panel

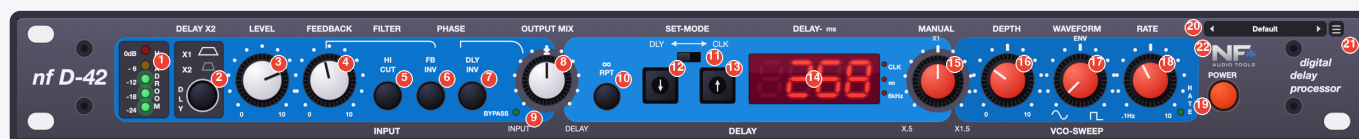


Figure 1 - Front panel. The numbers are explained on the following pages.

#	Control	What it does
1	HEADROOM	Five lamps (-24, -18, -12, -6, 0 dB) show the peak level seen by the converter after the LEVEL control and the input limiter.
2	DELAY X2	Selects the short range (X1, 16 kHz bandwidth) or the long range (X2, 6 kHz bandwidth, twice the delay time). The 6kHz lamp lights in X2.
3	LEVEL	Input gain, 0 to 10. 7.5 is unity gain (0 dB); 10 gives +20 dB.
4	FEEDBACK	Feeds the delayed signal back to the input, 0 to 10. At high settings the echoes sustain and become resonant.
5	HI CUT	Puts a 6 dB/octave low-pass filter (-3 dB at 4 kHz) in the feedback path: darker, more natural repeats.
6	FB INV	Inverts the polarity of the signal fed back. Changes the phase cancellations; typical for resonant flanging.
7	DLY INV	Inverts the polarity of the delayed sound sent to the output mix (flange notches become peaks).
8	OUTPUT MIX	Blend of direct and delayed sound. Fully left = direct only, fully right = delayed only. The centre detent is the equal blend for deepest flanging notches.

#	Control	What it does
9	BYPASS	Click the lamp or the word BYPASS: the delayed signal is removed from the output and nothing new enters the delay memory.
10	INFINITE REPEAT	Captures the sound in the delay memory at the next clock pulse and repeats it until the button is pressed again (see "Infinite repeat").
11	SET-MODE	DLY: the display shows the delay time and UP/DOWN change it. CLK: the display shows the clock fraction and UP/DOWN change it.
12	DOWN	DLY: shorter delay (one step). CLK: steps the numerator of the clock fraction through 1, 3, 5, 7, 9.
13	UP	DLY: longer delay (one step). CLK: steps the denominator of the clock fraction through 1, 2, 4, 8, 16, 32, 64.
14	DISPLAY	Four-digit read-out of the real delay time in milliseconds, following every sweep. In CLK mode: the clock fraction. Lamps: CLK (clock pulse), infinity (repeat mode), 6kHz (X2).
15	MANUAL	Manual VCO-sweep: changes the delay time by a factor of 0.5 (fully left) to 1.5 (fully right); the centre is X1. Use it for flanging by hand, pitch twisting and very long delays.
16	DEPTH	Amount of the internal sweep, 0 (none) to 10 (a full 3:1 range). At 10 the MANUAL control has no effect.
17	WAVEFORM	Fully left = sine, centre = envelope follower (the delay follows the input level), fully right = square. In between the shapes are blended.
18	RATE	Speed of the sweep, 0.1 Hz to 10 Hz. The lamp beside it flashes with the sweep. No effect in the envelope position.
19	POWER	Switches the unit on and off. Off = the signal passes untouched. At power-up the delay memory is empty, the repeat is off and the clock fraction is 1/2.
20	PRESET TAB	Arrows step through the presets; click the name for the list: Default, factory presets, your presets, Save and Load.
21	3-LINE MENU	Opens this manual (English or Portuguese) and the About box.
22	NF AUDIO TOOLS logo	Click the logo to return the plug-in window to its default size. The window can be resized with the corner handle; the size is remembered.

Delay times

The delay memory has 256 taps. The UP/DOWN buttons (or the display) choose the tap; the delay time depends on the tap, the DELAY X2 button and the MANUAL control:

Range	MANUAL = X0.5	MANUAL = X1	MANUAL = X1.5	Bandwidth	Step at X1
X1 (short)	400 ms	800 ms	1200 ms	16 kHz	3.1 ms
X2 (long)	800 ms	1600 ms	2400 ms	6 kHz	6.25 ms

Values are the maximum delay (tap 255) for each setting. The display always shows the actual time, including the sweep.

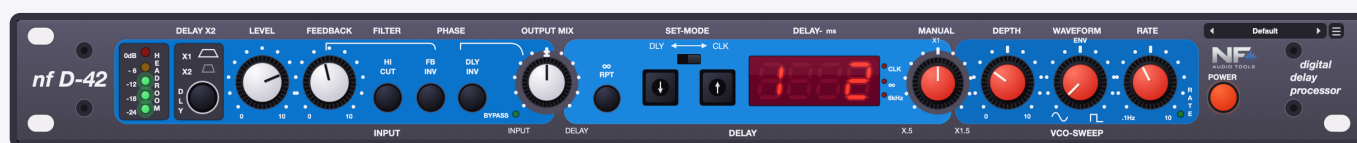
When you change the tap the delay glides for about 30 ms to avoid clicks.

Infinite repeat and the clock

Pressing INFINITE REPEAT does not freeze the sound at once: the capture happens at the next pulse of the clock (the CLK lamp flashes in time with it). From then on the whole delay memory is repeated until you press the button again, and nothing new is recorded. The loop length is the entire memory, so it does not depend on the delay tap, but it does change with MANUAL and DELAY X2 (for example 800 ms at X1 / short range, 2400 ms at X1.5 / long range). The direct signal still passes, so you can layer new parts over the loop (OUTPUT MIX). The plug-in never starts in repeat mode.

The clock period is the memory length times the clock fraction: with the memory at 800 ms and the fraction 1/2 the clock pulses every 400 ms. In SET-MODE CLK, DOWN steps the numerator (1, 3, 5, 7, 9) and UP the denominator (1, 2, 4, 8, 16, 32, 64). To catch a loop quickly, use a small fraction (for example 1/64); use a large fraction to wait for a precise moment.

The clock follows the sweep, so with DEPTH above 0 the pulses are not steady.



SET-MODE CLK: the display shows the clock fraction (numerator on the left, denominator on the right).

VCO sweep: flanger, chorus, vibrato

The sweep changes the speed of the delay memory clock, so the delay time moves between 0.5x and 1.5x of its value (a 3:1 range). DEPTH sets how much of that range the sweep uses, MANUAL sets where you are in it, RATE how fast it moves and WAVEFORM the shape: sine (smooth), square (the delay jumps), or the envelope follower (loud notes push the delay one way, quiet notes the other).

Short delays (a few milliseconds) with feedback give flanging; around 20 to 30 milliseconds with a slow sine gives chorus and doubling; with OUTPUT MIX fully right and a fast sine you get vibrato; a square wave with DEPTH raised gives pitch jumps; MANUAL turned slowly bends pitch like a tape machine.

Applications

Effect	How to set it
Echo / slap-back	Delay 100 to 500 ms, FEEDBACK 1 to 4, HI CUT on, OUTPUT MIX about 3 to 4, DEPTH 0.
Double tracking	Delay 20 to 40 ms, FEEDBACK 0, OUTPUT MIX centre, small DEPTH (1 to 2) with a slow RATE.
Flanging	Delay 3 to 10 ms (small tap, MANUAL left), OUTPUT MIX at the centre detent, FEEDBACK 4 to 6, DEPTH 10, RATE 1 to 3.
Resonant effects	As flanging with FEEDBACK 8 to 9, FB INV on and HI CUT on.
Vibrato	OUTPUT MIX fully right (delayed only), delay 10 to 20 ms, FEEDBACK 0, DEPTH 2 to 4, sine, RATE 6 to 7.
Pitch twisting	OUTPUT MIX fully right, DEPTH 6, square or sine; or turn MANUAL by hand while a sound plays.
Looping	Play a phrase, press INFINITE REPEAT, then layer over it. Use SET-MODE CLK to choose when the capture happens.

Presets

The preset tab has Default (every control at its starting position), 12 factory presets and a list of the presets you save. Save writes a file to Documents/NF Audio Tools/NF D-42/Presets.

Preset	Delay	Feedback	Mix	Switches	Depth	Wave	Rate
Slapback	100 ms	1.0	3.5	HI CUT	0.0	Sine	0.6 Hz
Tape-style Echo	300 ms	4.2	3.5	HI CUT	0.4	Sine	0.4 Hz
Long Echo	750 ms	5.0	3.2	HI CUT X2	0.0	Sine	0.6 Hz
Double Tracking	28 ms	0.0	5.0	-	1.2	Sine	0.4 Hz
Chorus	25 ms	0.5	5.0	HI CUT	3.5	Sine	0.5 Hz
Flanger	6 ms	5.0	5.0	-	10.0	Sine	0.3 Hz
Resonant Flange	6 ms	8.2	5.0	HI CUT FB INV	10.0	Sine	0.2 Hz
Through-Zero Style	9 ms	4.0	5.0	FB INV DLY INV	10.0	Sine	0.3 Hz
Vibrato	19 ms	0.0	10.0	-	3.0	Sine	1.9 Hz
Pitch Twist	250 ms	0.0	10.0	-	6.0	Square	0.4 Hz
Envelope Wobble	38 ms	3.0	5.0	HI CUT	6.0	Envelope	0.6 Hz
Looper (1/2 clock)	400 ms	0.0	5.0	-	0.0	Sine	0.6 Hz

Tips

- Keep an eye on the HEADROOM lamps: if the red lamp stays on, lower LEVEL. The input limiter is gentle by design, but it still colours the sound.
- Feedback at 10 sustains for a very long time. Lower it, switch HI CUT on, or use BYPASS to stop the repeats at once.
- If a tempo-based delay is needed, work out the time in milliseconds ($60000 / \text{BPM}$) and enter it with the display; the clock output of the hardware is represented by the CLK lamp.
- All controls can be automated in the host. The delay tap, clock numerator and denominator are stepped parameters.

Specifications

Item	Value
Formats	VST3, Audio Unit, AAX (macOS 10.15 or later, Apple Silicon and Intel)
Channels	Mono and stereo; both channels share the same sweep and clock
Delay range	X1: 0 to 1200 ms; X2: 0 to 2400 ms (see table)
Delay taps	256 per range
Sweep	Depth 0 to a full 3:1 range; rate 0.1 to 10 Hz; sine, envelope, square and blends
Bandwidth	16 kHz (X1) and 6 kHz (X2), -3 dB, anti-alias and reconstruction filters
Input stage	5:1 soft-knee compression above -3 dB, soft limiter at 0 dB
Feedback filter	6 dB/octave low-pass, -3 dB at 4 kHz
Latency	None (the direct signal is not delayed)

About

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